

DESALINATION: A POTENTIAL SOLUTION TO WATER SHORTAGE

The Ministry of Earth Sciences with the mission of developing economically viable technologies that use the Low Temperature Thermal Desalination (LTTD) based plants for desalination and renewable ocean resources for power has been successfully working on producing potable water through sustainable technology that can meet its goals while safeguarding its oceanic environment.

National Institute of Ocean Technology (NIOT), an autonomous body under the Ministry of Earth Sciences (MoES) is involved in developing technologies for ocean related activities such as deep sea mining, remotely operated submersibles and activities related to coastal and environmental protection. NIOT has also been focussing on the utilization of the ocean resources to find alternative technologies for fresh water and renewable energy. Currently the group is working on three specific areas, fresh water production using low temperature thermal desalination (LTTD) process, and energy production using two distinctly different processes, Ocean Thermal Energy Conversion and Wave Energy.

Desalination by LTTD process: The concept of the variation in the ocean water temperature with an increase in depth is used in the Low Temperature Thermal Desalination (LTTD) to flash evaporate the warm water at low pressure and condense the resulting vapour with the deep sea cold water. Starting from a small laboratory based 5m³/day capacity scale model, the group has successfully commissioned one 100m³/day capacity land based plant at Kavaratti and one 1000m³/day capacity floating barge based desalination plant off the coast of Chennai.

The group is also working on adapting the LTTD process for power plants where huge amounts of warm water are discharged continuously from the plant.

Subsequent to the success of the Kavaratti plant, the Lakshadweep Administration has approached NIOT to set-up similar island based plants in Andrott, Agatti and Minicoy.

Land based Desalination Plant at Kavaratti: The plant that started in July 2005 as a pilot project to demonstrate the potential of the LTTD process has now become the backbone for the Kavaratti water distribution system that pumps the water daily to the local public. Since its inception, the plant was handled by NIOT until March 2006 and then handed over to local administration. The plant has crossed 8 crore liters of production.

Barge Mounted Desalination Plant off Tuticorin: An experimental 50m³ barge mounted LTTD plant was successfully made operational off Tuticorin in March 2005 and this paved the way for the larger capacity plant.

Barge Mounted Desalination Plant off Chennai: A floating Barge Mounted 1000 m³/day Desalination plant that can cater to the needs of mainland was commissioned 40km off Chennai coast in April 2007. The experimental plant was operational for a few weeks. In the process of this demonstration, NIOT has studied various configurations for the cold water pipe and the connection between the pipe and the vessel. NIOT is now working on higher capacity plants for mainland needs.

